

Propel

AI-Powered CV Generator

Microservices · Event-Driven · Intelligent Agents

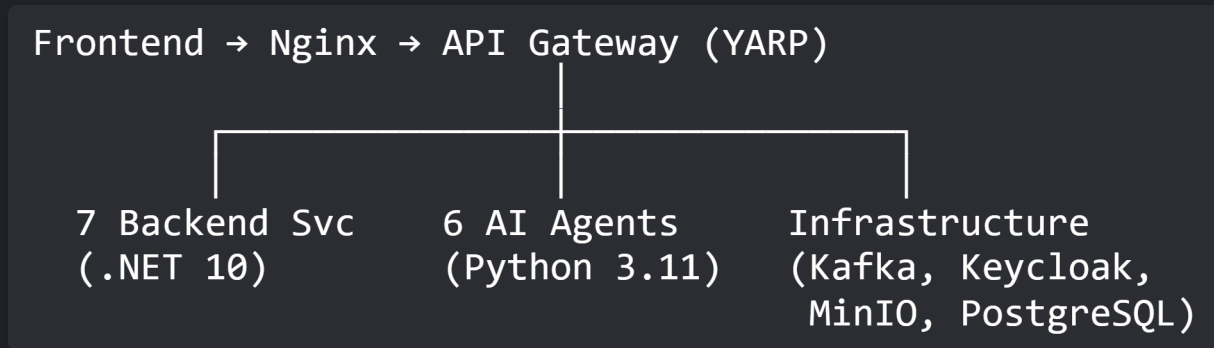
What is Propel?

- AI-driven CV generation from job descriptions
- Job application tracking with Kanban & analytics
- Portfolio management (projects, skills, experience)
- Live job offer aggregation via crawling
- Email notifications & scheduled reminders
- Keycloak SSO with Google/GitHub providers

Tech Stack

Layer	Technology
Frontend	Angular 21, TypeScript, RxJS, Signals, SCSS
Backend	.NET 10, ASP.NET Core, EF Core, gRPC
AI Agents	Python 3.11, FastAPI, LangChain, RAG
Database	PostgreSQL 16 + pgvector (8 databases)
Auth	Keycloak 26.2, OpenID Connect, JWT
Messaging	Confluent Kafka (KRaft mode)
Storage	MinIO (S3-compatible)
Container	Docker, Docker Compose, Nginx

System Architecture



“ Full diagram: `docs/diagrams/system-architecture.eraserdiagram` ”

Backend Services (.NET 10)

- **User Service** — CRUD, Keycloak sync, gRPC
- **User Content** — Portfolio (projects, skills, experience, education, etc.)
- **Workflow** — CV generation orchestration
- **Application** — Job tracking, Kanban, analytics
- **CV Service** — CV creation, versioning, PDF/DOCX
- **Job Offer** — Live crawling, SignalR real-time
- **Notification** — Email, reminders, Hangfire

AI Agent Pipeline

Flow: Submit JD → Extract → Match → Render →
Optimize → Deliver

1. **Job Extractor** — Parses JD → structured requirements
2. **Search Agent** — RAG matching via pgvector
3. **Template Agent** — Renders CV with Jinja2/LaTeX
4. **CV Optimizer** — ATS compliance, keywords
5. **Contact Agent** — Email delivery + cover letter
6. **Job Crawler** — Automated job board crawling

Full diagram: `docs/diagrams/cv-pipeline.eraserdiagram`

Deployment — 8 VMs / 5 Team Members

Owner	VM	Services
Bilal	VM1	Frontend + API Gateway + Monitoring
Bilal	VM2	User + Content + Application + Notification
Bilal	VM3	Extractor + Contact + Template agents
Owner	VM	Services
Somia	VM1	Workflow Service
Somia	VM2	Optimizer + Search + Crawler agents
Nourelhouda	VM1	Keycloak + Kafka + MinIO + DBs
Mouhssine	VM1	CV + Job Offer services
Salman	VM1	Prometheus + Grafana

Use Cases

Diagram	File
Authentication	<code>docs/diagrams/use-cases/authentication.puml</code>
Profile Management	<code>docs/diagrams/use-cases/profile-management.puml</code>
CV Generation	<code>docs/diagrams/use-cases/cv-generation.puml</code>
Application Tracking	<code>docs/diagrams/use-cases/application-tracking.puml</code>
Jobs & Notifications	<code>docs/diagrams/use-cases/jobs-notifications.puml</code>

All diagrams generated with **PlantUML**.

Kafka Event Flow

```
API Gateway —> user.registered —> Notification Service  
                      (userId, email, username)
```

```
Job Offer Service —> trigger-live-crawl —> Job Crawler  
                      (keywords, location)
```



Full diagram: `docs/diagrams/event-flow.eraserdiagram`



Key Features

- ✨ **AI CV Generation** — Tailored CV in 11 seconds
- 📊 **Kanban Dashboard** — Drag & drop application tracking
- 📈 **Analytics** — Response rates, trends, statistics
- 🔔 **Smart Reminders** — Follow-up scheduling
- 🔒 **SSO** — Keycloak with Google/GitHub
- 🔄 **Real-time** — Kafka events + SignalR updates
- 🐳 **Containerized** — Full Docker Compose deployment

Thank You

Propel — AI-Powered CV Generator

Diagrams · Slides

Questions?